

Week 7 - Skill

Part 1 — Process Synchronization and Child Process Monitoring using waitpid()

1. Creating a working folder

```
samanth-ubuntu@samanth-ubuntu:~/week6_s$ mkdir -p ~/week7_s
cd ~/week7_s
pwd
/home/samanth-ubuntu/week7_s
```

2. Create the C program and Compile the program

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ nano waitpid_demo.c
samanth-ubuntu@samanth-ubuntu:~/week7_s$ gcc waitpid_demo.c -o waitpid_demo
ls -l waitpid_demo
-rwxrwxr-x 1 samanth-ubuntu samanth-ubuntu 70856 Oct  9 08:19 waitpid_demo
```

3. Execute the program

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ ./waitpid_demo
Parent: PID = 7373
Parent: Waiting for child PID = 7374
Child: PID = 7374
Child: Performing some work...
Child: Work completed. Exiting...
Parent: Child PID 7374 has terminated.
Parent: Child exited normally with status 5
```

4. Demonstrate witpid() behaviour

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ ./waitpid_demo
Parent: PID = 7383
Parent: Waiting for child PID = 7384
Child: PID = 7384
Child: Performing some work...
Child: Work completed. Exiting...
Parent: Child PID 7384 has terminated.
Parent: Child exited normally with status 5
```

Summary:

The program successfully created a child process using fork(). The parent process used waitpid() to wait for the child to complete and then retrieved the child's exit status, demonstrating process synchronization and child process monitoring.

PART 2 — PATH Variable & Command Resolution

1. Checking your PATH

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ echo $PATH
which ls
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/bin/ls
```

2. Create the PATH resolver program

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ nano path_resolver.c
samanth-ubuntu@samanth-ubuntu:~/week7_s$ gcc path_resolver.c -o path_resolver
ls -l path_resolver
-rwxrwxr-x 1 samanth-ubuntu samanth-ubuntu 70928 Oct  9 08:27 path_resolver
```

3. Test command resolution

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ ./path_resolver ls
Command to resolve: ls
Searching PATH directories...

Executable found: /usr/bin/ls
Permission: Executable
```

4. Test a missing command

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ ./path_resolver abcxyz
Command to resolve: abcxyz
Searching PATH directories...

Command not found in PATH.
```

5. Test another real command

```
samanth-ubuntu@samanth-ubuntu:~/week7_s$ ./path_resolver gcc
Command to resolve: gcc
Searching PATH directories...

Executable found: /usr/bin/gcc
Permission: Executable
```

Summary:

The program successfully retrieved the PATH variable and searched its directories to locate executable commands. It verified executable permissions and correctly reported when a requested command was not found.